In [1]:
```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
```

In [2]:
```
iris = pd.read_csv("C:/Users/Ganesh/Downloads/Iris.csv")
```

In [3]:
```
iris.head()
```

Out[3]:

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Species
0	1	5.1	3.5	1.4	0.2	Iris-setosa
1	2	4.9	3.0	1.4	0.2	Iris-setosa
2	3	4.7	3.2	1.3	0.2	Iris-setosa
3	4	4.6	3.1	1.5	0.2	Iris-setosa
4	5	5.0	3.6	1.4	0.2	Iris-setosa

In [4]:
```
iris.tail()
```

Out[4]:

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Species
145	146	6.7	3.0	5.2	2.3	Iris-virginica
146	147	6.3	2.5	5.0	1.9	Iris-virginica
147	148	6.5	3.0	5.2	2.0	Iris-virginica
148	149	6.2	3.4	5.4	2.3	Iris-virginica
149	150	5.9	3.0	5.1	1.8	Iris-virginica

In [5]:
```
iris.shape
```

Out[5]:
```
(150, 6)
```

In [6]:
```
print(iris.columns)
```

```
Index(['Id', 'SepalLengthCm', 'SepalWidthCm', 'PetalLengthCm', 'PetalWidthCm',
       'Species'],
      dtype='object')
```

In [7]:
```
iris.isnull().sum()
```

Out[7]:
```
Id               0
SepalLengthCm    0
SepalWidthCm     0
PetalLengthCm    0
PetalWidthCm     0
Species          0
dtype: int64
```

In [8]:
```
iris.describe()
```

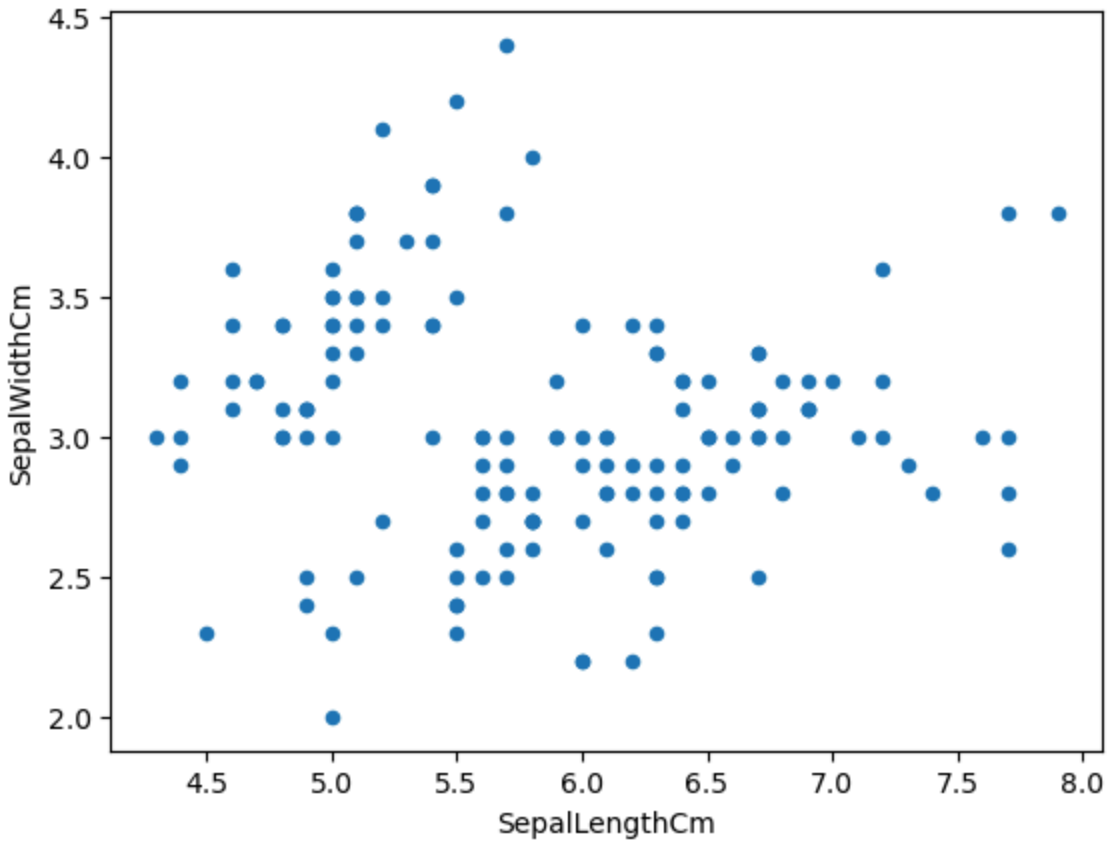
Out[8]:

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm
count	150.000000	150.000000	150.000000	150.000000	150.000000
mean	75.500000	5.843333	3.054000	3.758667	1.198667
std	43.445368	0.828066	0.433594	1.764420	0.763161
min	1.000000	4.300000	2.000000	1.000000	0.100000
25%	38.250000	5.100000	2.800000	1.600000	0.300000
50%	75.500000	5.800000	3.000000	4.350000	1.300000
75%	112.750000	6.400000	3.300000	5.100000	1.800000
max	150.000000	7.900000	4.400000	6.900000	2.500000

In [9]:
```
iris['Species'].value_counts()
```

Out[9]:
```
Species
Iris-setosa        50
Iris-versicolor    50
Iris-virginica     50
Name: count, dtype: int64
```

In [12]:
```
iris.plot(kind='scatter', x='SepalLengthCm',
y='SepalWidthCm')
plt.show()
```



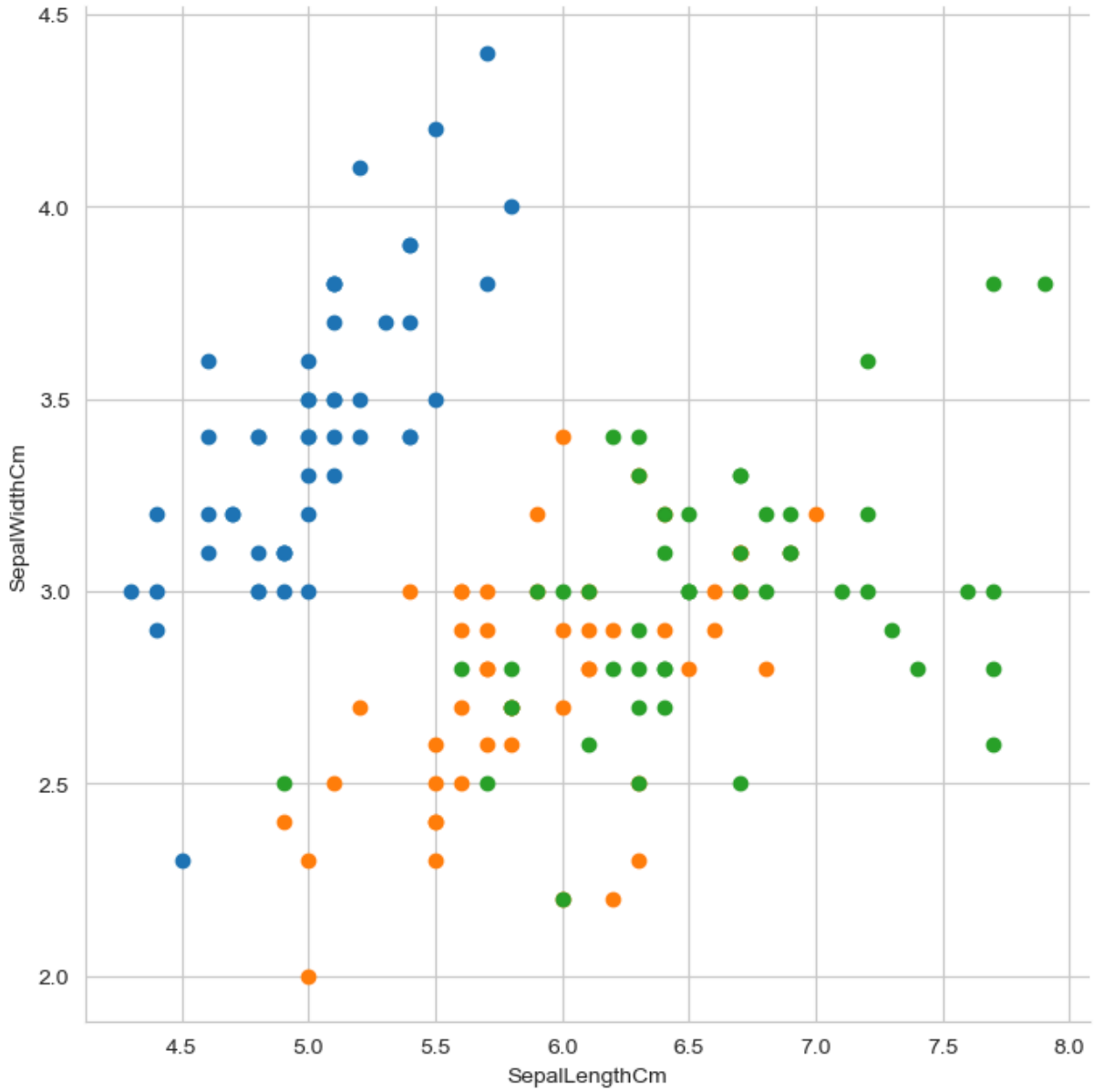
In [15]:
```
iris_numeric.mean()
```

Out[15]:
```
Id               75.500000
SepalLengthCm     5.843333
SepalWidthCm      3.054000
PetalLengthCm     3.758667
PetalWidthCm      1.198667
dtype: float64
```

In [16]:
```
iris.Species.mode()
```

Out[16]:
```
0        Iris-setosa
1    Iris-versicolor
2     Iris-virginica
Name: Species, dtype: object
```

In [22]:
```
import seaborn as sns
sns.set_style("whitegrid")
sns.FacetGrid(iris, hue="Species", height=7).map(plt.scatter, "SepalLengthCm", "Sep
plt.show()
```



In [23]:
```
setosa_data = iris[iris['Species'] == 'Iris-setosa']
versicolor_data = iris[iris['Species'] == 'Iris-versicolor']
```

In [24]:
```
setosa_data
```

Out[24]:

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Species
0	1	5.1	3.5	1.4	0.2	Iris-setosa
1	2	4.9	3.0	1.4	0.2	Iris-setosa
2	3	4.7	3.2	1.3	0.2	Iris-setosa
3	4	4.6	3.1	1.5	0.2	Iris-setosa
4	5	5.0	3.6	1.4	0.2	Iris-setosa
5	6	5.4	3.9	1.7	0.4	Iris-setosa
6	7	4.6	3.4	1.4	0.3	Iris-setosa
7	8	5.0	3.4	1.5	0.2	Iris-setosa
8	9	4.4	2.9	1.4	0.2	Iris-setosa
9	10	4.9	3.1	1.5	0.1	Iris-setosa
10	11	5.4	3.7	1.5	0.2	Iris-setosa
11	12	4.8	3.4	1.6	0.2	Iris-setosa
12	13	4.8	3.0	1.4	0.1	Iris-setosa
13	14	4.3	3.0	1.1	0.1	Iris-setosa
14	15	5.8	4.0	1.2	0.2	Iris-setosa
15	16	5.7	4.4	1.5	0.4	Iris-setosa
16	17	5.4	3.9	1.3	0.4	Iris-setosa
17	18	5.1	3.5	1.4	0.3	Iris-setosa
18	19	5.7	3.8	1.7	0.3	Iris-setosa
19	20	5.1	3.8	1.5	0.3	Iris-setosa
20	21	5.4	3.4	1.7	0.2	Iris-setosa
21	22	5.1	3.7	1.5	0.4	Iris-setosa
22	23	4.6	3.6	1.0	0.2	Iris-setosa
23	24	5.1	3.3	1.7	0.5	Iris-setosa
24	25	4.8	3.4	1.9	0.2	Iris-setosa
25	26	5.0	3.0	1.6	0.2	Iris-setosa
26	27	5.0	3.4	1.6	0.4	Iris-setosa
27	28	5.2	3.5	1.5	0.2	Iris-setosa
28	29	5.2	3.4	1.4	0.2	Iris-setosa
29	30	4.7	3.2	1.6	0.2	Iris-setosa

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Species
30	31	4.8	3.1	1.6	0.2	Iris-setosa
31	32	5.4	3.4	1.5	0.4	Iris-setosa
32	33	5.2	4.1	1.5	0.1	Iris-setosa
33	34	5.5	4.2	1.4	0.2	Iris-setosa
34	35	4.9	3.1	1.5	0.1	Iris-setosa
35	36	5.0	3.2	1.2	0.2	Iris-setosa
36	37	5.5	3.5	1.3	0.2	Iris-setosa
37	38	4.9	3.1	1.5	0.1	Iris-setosa
38	39	4.4	3.0	1.3	0.2	Iris-setosa
39	40	5.1	3.4	1.5	0.2	Iris-setosa
40	41	5.0	3.5	1.3	0.3	Iris-setosa
41	42	4.5	2.3	1.3	0.3	Iris-setosa
42	43	4.4	3.2	1.3	0.2	Iris-setosa
43	44	5.0	3.5	1.6	0.6	Iris-setosa
44	45	5.1	3.8	1.9	0.4	Iris-setosa
45	46	4.8	3.0	1.4	0.3	Iris-setosa
46	47	5.1	3.8	1.6	0.2	Iris-setosa
47	48	4.6	3.2	1.4	0.2	Iris-setosa
48	49	5.3	3.7	1.5	0.2	Iris-setosa
49	50	5.0	3.3	1.4	0.2	Iris-setosa

In [25]:
```
versicolor_data
```

Out[25]:

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Species
50	51	7.0	3.2	4.7	1.4	Iris-versicolor
51	52	6.4	3.2	4.5	1.5	Iris-versicolor
52	53	6.9	3.1	4.9	1.5	Iris-versicolor
53	54	5.5	2.3	4.0	1.3	Iris-versicolor
54	55	6.5	2.8	4.6	1.5	Iris-versicolor
55	56	5.7	2.8	4.5	1.3	Iris-versicolor
56	57	6.3	3.3	4.7	1.6	Iris-versicolor
57	58	4.9	2.4	3.3	1.0	Iris-versicolor
58	59	6.6	2.9	4.6	1.3	Iris-versicolor
59	60	5.2	2.7	3.9	1.4	Iris-versicolor
60	61	5.0	2.0	3.5	1.0	Iris-versicolor
61	62	5.9	3.0	4.2	1.5	Iris-versicolor
62	63	6.0	2.2	4.0	1.0	Iris-versicolor
63	64	6.1	2.9	4.7	1.4	Iris-versicolor
64	65	5.6	2.9	3.6	1.3	Iris-versicolor
65	66	6.7	3.1	4.4	1.4	Iris-versicolor
66	67	5.6	3.0	4.5	1.5	Iris-versicolor
67	68	5.8	2.7	4.1	1.0	Iris-versicolor
68	69	6.2	2.2	4.5	1.5	Iris-versicolor
69	70	5.6	2.5	3.9	1.1	Iris-versicolor
70	71	5.9	3.2	4.8	1.8	Iris-versicolor
71	72	6.1	2.8	4.0	1.3	Iris-versicolor
72	73	6.3	2.5	4.9	1.5	Iris-versicolor
73	74	6.1	2.8	4.7	1.2	Iris-versicolor
74	75	6.4	2.9	4.3	1.3	Iris-versicolor
75	76	6.6	3.0	4.4	1.4	Iris-versicolor
76	77	6.8	2.8	4.8	1.4	Iris-versicolor
77	78	6.7	3.0	5.0	1.7	Iris-versicolor
78	79	6.0	2.9	4.5	1.5	Iris-versicolor
79	80	5.7	2.6	3.5	1.0	Iris-versicolor

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Species
80	81	5.5	2.4	3.8	1.1	Iris-versicolor
81	82	5.5	2.4	3.7	1.0	Iris-versicolor
82	83	5.8	2.7	3.9	1.2	Iris-versicolor
83	84	6.0	2.7	5.1	1.6	Iris-versicolor
84	85	5.4	3.0	4.5	1.5	Iris-versicolor
85	86	6.0	3.4	4.5	1.6	Iris-versicolor
86	87	6.7	3.1	4.7	1.5	Iris-versicolor
87	88	6.3	2.3	4.4	1.3	Iris-versicolor
88	89	5.6	3.0	4.1	1.3	Iris-versicolor
89	90	5.5	2.5	4.0	1.3	Iris-versicolor
90	91	5.5	2.6	4.4	1.2	Iris-versicolor
91	92	6.1	3.0	4.6	1.4	Iris-versicolor
92	93	5.8	2.6	4.0	1.2	Iris-versicolor
93	94	5.0	2.3	3.3	1.0	Iris-versicolor
94	95	5.6	2.7	4.2	1.3	Iris-versicolor
95	96	5.7	3.0	4.2	1.2	Iris-versicolor
96	97	5.7	2.9	4.2	1.3	Iris-versicolor
97	98	6.2	2.9	4.3	1.3	Iris-versicolor
98	99	5.1	2.5	3.0	1.1	Iris-versicolor
99	100	5.7	2.8	4.1	1.3	Iris-versicolor

In [26]:
```
setosa_stats = {
'percentile': np.percentile(setosa_data['SepalLengthCm'],
50), 'mean': np.mean(setosa_data['SepalLengthCm']),
'std_dev': np.std(setosa_data['SepalLengthCm'])
}
```

In [27]:
```
versicolor_stats = {
'percentile': np.percentile(versicolor_data['SepalLengthCm'],
50), 'mean': np.mean(versicolor_data['SepalLengthCm']),
'std_dev': np.std(versicolor_data['SepalLengthCm'])
}
```

In [29]:
```
print('Statistics for Iris Setosa:')
print(setosa_stats)
print('')
```

```
print('Statistics for Iris Versicolor:')
print(versicolor_stats)
```

```
Statistics for Iris Setosa:
{'percentile': np.float64(5.0), 'mean': np.float64(5.006), 'std_dev': 0.348946987377
739}

Statistics for Iris Versicolor:
{'percentile': np.float64(5.9), 'mean': np.float64(5.936), 'std_dev': 0.510983365678
3752}
```

In [ ]: